

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 5343A	Course Title: Natural Language Processing
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Learn the techniques in natural language processing.
- Familiar with the natural language generation.
- Understand machine translation and speech recognition.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of AI	3344	Artificial intelligence	III
Basic of ML	4341	Mathematics for ML	IV

Course Outcomes :

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Summarise the concepts of NLP and Language Modelling	16	Understanding
CO2	Outline Word Level Analysis and Syntactic Analysis	15	Understanding
CO3	Explain Semantic Analysis and Discourse Processing	12	Understanding
CO4	Summarize Natural Language Generation and Machine Translation	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level
CO1	Summarise the concepts of NLP and Language Modelling		
M1.01	Describe the concepts of NLP	1	Understanding
M1.02	Explain NLP Pipeline	1	Understanding
M1.03	Summarise the Phases of NLP	2	Understanding
M1.04	Outline the challenges of NLP	2	Understanding
M1.05	Describe language and grammar	3	Understanding
M1.06	Explain various Grammar-based Language Model	4	Understanding
M1.07	Summarise Statistical Language Model	3	Understanding
Contents : Concepts: Introduction to Natural Language Processing (NLP), components of NLP , Applications Language and Knowledge -NLP Pipeline, Phases of NLP, The Challenges of NLP, Language and Grammar Language Modeling: Various Grammar-based Language Models, Statistical Language Model			
CO2	Outline Word Level Analysis and Syntactic Analysis		
M2.01	Summarise the properties of Regular Expressions	2	Understanding
M2.02	Outline Finite-State Automata	3	Understanding
M2.03	Explain Morphological Parsing	3	Understanding
M2.04	Outline Spelling Error Detection and Correction	2	Understanding

M2.05	Describe Part-of Speech Tagging	3	Understanding
M2.06	Summarise Syntactic Analysis	4	Understanding
	Series Test -1	1	
Contents: Preprocessing of text -Tokenization, Filtration, Script Validation, Stop Word Removal, Stemming, Lemmatization Word Level Analysis: Introduction, Regular Expressions, Properties, Finite-State Automata- Deterministic, Non-Deterministic, Morphological Parsing, Spelling Error Detection and Correction, Words and Word Classes, Part-of-Speech Tagging. Syntactic Analysis: Context-Free Grammar, Parsing, Probabilistic Parsing			
CO3	Explain Semantic Analysis and Discourse Processing		
M3.01	Explain Semantic Analysis	3	Understanding
M3.02	Describe the approaches and methods of Word Sense Disambiguation	3	Understanding
M3.03	Summarize Discourse Processing	3	Understanding
M3.04	Outline Discourse Coherence and Structure	3	Understanding
Contents: Semantic Analysis: Meaning Representation, Lexical Semantics, Ambiguity, Word Sense Disambiguation. Discourse Processing: Cohesion, Reference Resolution, Discourse Coherence and Structure.			
CO4	Summarize Natural Language Generation(NLG) and Machine Translation		
M4.01	Explain the architecture of NLG Systems, Generation Tasks and Representations	3	Understanding
M4.02	Describe Machine Translation and its problems	3	Understanding
M4.03	Explain Machine Translation Approaches	2	Understanding
M4.04	Summarise the pipeline used for speech recognition	2	Understanding
M4.05	Outline the steps for processing and extracting features from sound signals	4	Understanding
	Series Test -2	1	
Contents: Introduction-Architecture of Natural Language Generation (NLG) Systems, Generation Tasks and Representations-Application of NLG. Machine Translation, Problems in Machine Translation, Machine Translation Approaches Speech Recognition: Learn the pipeline used for speech recognition. Learn to process and extract features from sound signals.			

Text / Reference

T/R	Book Title/Author
T1	Tanveer Siddiqui, U.S. Tiwary, “Natural Language Processing and Information Retrieval”, Oxford University Press, 2008.
T2	Daniel Jurafsky and James H Martin, “Speech and Language Processing: An introduction to Natural Language Processing, Computational Linguistics and Speech Recognition”, 2 nd Edition, Prentice Hall, 2008.
T3	James Allen, “Natural Language Understanding”, 2 nd edition, Benjamin /Cummings publishing company, 1995.

Online Resources

Sl.No	Website Link
1	http://www.coursera.org/specializations/natural-language-processing
2	https://www.deeplearning.ai/resources/natural-language-processing